# C Programming Assignment – Question 6

## 6. Car Race

## Question

Three roads A, B, and C are assigned numbers, followed by the number of cars. Determine the road the car will take. If no road is possible, print “No path exists”. The selection rule is not fully specified in the assignment.

## Sample Input

```
243
136
567
17
```

## Sample Output

```
Car 1 goes into road B
```

## C Program

```
#include <stdio.h>

int main(void) {
    int A, B, C, cars;

    printf("Enter numbers for roads A, B and C: ");
    scanf("%d%d%d", &A, &B, &C);

    printf("Enter number of cars: ");
    scanf("%d", &cars);

    /* Assumption: choose the first road whose number
       is divisible by the number of cars. */
    if (cars != 0 && A % cars == 0) {
        printf("Car 1 goes into road A\n");
    } else if (cars != 0 && B % cars == 0) {
        printf("Car 1 goes into road B\n");
    } else if (cars != 0 && C % cars == 0) {
        printf("Car 1 goes into road C\n");
    } else {
        printf("No path exists\n");
    }

    return 0;
}
```

## Explanation

1. Read the numbers assigned to roads A, B, and C, plus the number of cars.

2. Check whether each road number is divisible by the number of cars.

3. Select the first matching road in A, B, C order.

4. If none matches, print “No path exists”.

## Note

Assumption used to match the sample: test divisibility by the number of cars. Since 136 is divisible by 17, Road B is selected.